



LITERATURE REVIEW

Representation Learning for Autonomous Peak Detection in Mass Spectrometry Imaging

Zayd SULIMAN
2653934

School of Computer Science and Applied Mathematics
Faculty of Science
University of the Witwatersrand

Supervisor:
Hairong Bau and Richard Klein

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Declaration of Authorship

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Chapter 1

Introduction

The characterization of biological tissues has historically relied on classical histological staining and light microscopy, requiring anatomical pathologists to diagnose diseases based on structural and morphological cellular alterations. However, the paradigm of modern precision medicine necessitates a significantly deeper understanding of the molecular pathogenesis underlying these morphological changes. To bridge the gap between traditional histology and high-throughput molecular profiling, Mass Spectrometry Imaging (MSI), particularly Matrix-Assisted Laser Desorption/Ionization (MALDI), has emerged as a revolutionary molecular microscopy technique over the past three decades. [Schwamborn and Caprioli 2010] By mapping the spatial distribution of thousands of distinct biomolecules ranging from small metabolites, lipids, and exogenous drugs to large intact proteins directly within an intact tissue section, MSI provides a label-free, highly multiplexed view of the cellular microenvironment. [Ye et al. 2013]

The ability to simultaneously capture spatial relationships and complex spectral data without prior knowledge of the target molecules has propelled MSI to the forefront of contemporary biomedical research, facilitating advancements in biomarker discovery, pharmaceutical drug mapping, and clinical diagnostics [Abdelmoula et al. 2021]. In complex localized pathologies such as glioblastoma, colorectal carcinoma, and infectious diseases, MSI enables the direct visualization of tumor heterogeneity, xenobiotic toxins, and localized immune responses that are otherwise entirely invisible to standard hematoxylin and eosin (H&E) staining. Despite these profound advantages and the rapid maturation of the hardware, the clinical translation of MSI into routine diagnostic pipelines is heavily impeded by massive computational

bottlenecks. A single high-resolution MSI experiment can generate three-dimensional data hypercubes ranging from tens of gigabytes to several terabytes per tissue slice [Körber et al. 2025]. Each individual pixel within this spatial grid contains a high-dimensional mass spectrum, often consisting of tens to hundreds of thousands of discrete mass-to-charge (m/z) bins [Abdelmoula et al. 2021]. Extracting biologically meaningful, statistically robust signals from this immense, noisy, and highly sparse data structure represents one of the most significant challenges in modern computational biology and medical image analysis [Zhang et al. 2025].

1.1 Problem Statement

Ideally, the computational analysis of clinical Mass Spectrometry Imaging datasets would be entirely automated, preserving the full spectrum of molecular information while seamlessly translating high-dimensional raw physical data into robust, clinically actionable diagnostic signatures. In such an ideal computational environment, machine learning models would accurately capture the highly non-linear relationships between thousands of interacting metabolites and proteins without requiring human bias or manual parameter intervention. This idealized pipeline would enable the immediate, unhindered identification of rare, low-abundance molecular biomarkers that are critical for the early detection and subtyping of complex diseases.

The reality of current MSI analysis, however, stands in stark contrast to this ideal. The conventional computational processing of MSI data is fundamentally limited by highly heuristic, multistep preprocessing pipelines that are required to compress the massive data hypercubes before any statistical analysis can occur. The most critical, computationally expensive, and subjective of these steps is "peak picking". Classical peak picking algorithms, such as those relying on sparse frame multipliers, continuous wavelet transforms, local maxima detection, or strict signal-to-noise ratio (SNR) thresholding, require extensive empirical parameter tuning [Lieb et al. 2020]. This subjectivity introduces a severe analytical bottleneck into the diagnostic pipeline: arbitrary thresholds inevitably discard low-abundance molecular signals, permanently masking crucial biological phenomena, while simultaneously retaining highly structured chemical noise derived from the matrix application process. Furthermore, traditional peak picking fundamentally destroys the underlying non-linear spectral manifold of the data, forcing all downstream

machine learning classifiers and clustering algorithms to operate on a highly biased, artificially truncated, and linear representation of the tissue microenvironment. [Abdelmoula et al. 2021]

To propose an improvement to this reality and transition the field toward the ideal analytical environment, it is proposed that unsupervised deep learning models be deployed directly on raw, un-preprocessed MSI data arrays. Artificial neural networks, specifically deep generative models such as Variational Autoencoders (VAEs) and contrastive representation learning networks, possess the mathematical capacity to learn the generative physical processes and spatial correlations of molecular ions without requiring heuristic thresholding or prior baseline correction [Abdelmoula et al. 2021]. By reconceptualizing peak detection not as an arbitrary signal-filtering task, but as an advanced manifold learning and spatial representation problem (often termed "peak learning") deep neural networks can autonomously discover biologically relevant m/z features while retaining the complex, non-linear structural relationships inherent in the raw mass spectrometry data. Developing, evaluating, and optimizing these advanced deep learning frameworks, particularly for highly diverse and noisy datasets sourced from local clinical hospital environments, forms the core research imperative of this study. [Guo et al. 2024]

1.2 Expected Contributions

This research aims to contribute a practical computational bridge to the field of spatial proteomics. By mathematically integrating physical neighborhood covariance into an existing, lightweight VAE framework, this project provides an optimized feature extraction pipeline. It evaluates whether we can achieve spatially-aware biochemical mapping and eliminate subjective human bias from MSI preprocessing without taking on the crippling computational burden of 3D convolutional networks.

Chapter 2

Background and Related Work

2.1 Background: Foundations of MSI

Mass Spectrometry Imaging represents a profound technological intersection between advanced analytical chemistry, physics, and spatial biology. The technology functions by systematically directing a highly focused ionization source across a meticulously defined two-dimensional grid superimposed on a thin biological tissue section. In the MALDI MSI workflow, the tissue is coated with an energy-absorbing organic matrix. A pulsed laser is fired at discrete coordinates, sublimating the matrix and inducing localized desorption and ionization of the endogenous tissue molecules. An analyzer then separates the ions based entirely on their mass-to-charge ratio (m/z).[\[Caprioli et al. 1997; Körber et al. 2025\]](#)

The resulting computational output is an exceptionally high-dimensional data hypercube. For a tissue section imaged at a specific spatial resolution yielding $N \times M$ spatial pixels, each individual pixel contains a complete mass spectrum. This structural complexity creates a profound "curse of dimensionality," where the sheer volume of data rapidly exceeds the capacities of standard analytical workstations. [\[Abdelmoula et al. 2021; Aspect Analytics 2026\]](#)

Due to the immense size, sparsity, and inherent noise of the raw spectral hypercube, classical workflows dictate the application of a heuristic-driven pre-processing pipeline (baseline subtraction, spectral smoothing, intensity normalization, and peak picking) prior to any downstream analysis. Peak picking, the most critical dimensionality reduction step, searches for local maxima that satisfy rigidly defined geometric criteria or SNR thresholds. However, this process relies on strictly linear assumptions. In clinical molecular

pathology, critical biomarkers often exist in exceptionally low endogenous abundances. When heuristic SNR thresholds are applied, these vital low-abundance ions are indiscriminately discarded. This severe limitation has driven the field to seek "peak learning" paradigms, wherein the data mathematically dictates feature extraction without arbitrary thresholds. [Abdelmoula et al. 2021; Buchberger et al. 2018]

2.2 Related Work: Deep Learning in MSI

The integration of deep learning neural networks has fundamentally altered the approach to high-dimensional medical image analysis. By leveraging multi-layered neural architectures, deep learning models can automatically extract latent features, accurately model spectrum-structure correlations, and construct low-dimensional spatial representations that preserve critical biological variance while discarding true stochastic noise.

Unsupervised Peak Learning via Generative Probabilistic Models

One of the most significant advancements in addressing the subjectivity of MSI preprocessing is the development of generative probabilistic models. The seminal work by Abdelmoula et al. [2021] introduced "msiPL" (Mass Spectrometry Imaging Peak Learning), a highly sophisticated framework leveraging a fully connected Variational Autoencoder (VAE) to perform unsupervised analysis directly on raw, un-preprocessed MSI datasets.

Unlike conventional algorithms that blindly search for local geometric maxima, msiPL treats the recorded mass spectrum as the output of a complex generative process. The encoder maps the high-dimensional raw spectrum into a lower-dimensional latent space vector, computing the mean and variance parameters that define this distribution. The decoder then attempts to perfectly reconstruct the original spectrum utilizing solely the compressed information.

The paradigm-shifting innovation of msiPL lies in how it executes "peak picking" after network optimization. Rather than applying post-hoc numerical thresholds, msiPL performs backpropagated threshold analysis directly on the learned weight parameters of the neural network. The structural

mathematical weights inherently encode the most critical m/z features required for accurate spectral reconstruction. The clinical efficacy of msiPL has been validated on complex datasets (e.g., glioblastoma, colorectal carcinomas), consistently identifying biologically significant m/z peaks that are routinely discarded by classical algorithms. [Abdelmoula et al. 2021 2022]

Spatially Aware Self-Supervised Peak Learning

While msiPL revolutionized spectral analysis, it evaluates mass-to-charge ratios without considering their spatial context across the tissue grid. Biological molecular signals inherently exhibit strong spatial coherence, whereas noise is typically random. To address this, Weigand et al. [2026] introduced the Spatial Self-Supervised Peak Learning (S3PL) framework.

Building upon foundational autoencoder concepts, S3PL is an end-to-end neural network that performs context-aware peak picking. It employs a 3D convolutional autoencoder that evaluates multi-dimensional spectral patches ($x \in \mathbb{R}^{h \times w \times c}$). The network learns to generate a continuous spatial attention mask based on local spatial coherence, actively penalizing the network's loss function if it attempts to reconstruct non-spatial, stochastic noise. By intrinsically linking peak learning to spatial structure, S3PL isolates highly refined, spatially coherent molecular signals, generating the ideal inputs required for downstream image-based models.

Self-Supervised Spatial Representation and Contrastive Learning

Extracting robust spatial relationships from isolated ion images requires specialized computer vision architectures. Accurately identifying colocalized ions and isotope ions is critical for biomarker discovery. Guo et al. [2024] introduced "DeepION," a deep learning-based low-dimensional representation model engineered for analyzing mass spectrometry ion images via a self-supervised contrastive learning framework, eliminating the need for manually annotated training data.

DeepION is built upon the SimSiam contrastive model, utilizing twin neural networks with a ResNet18 convolutional backbone. To prevent total model collapse, a prediction multilayer perceptron (MLP) is utilized, and dimensionality reduction via UMAP condenses the representation vector into a 20-dimensional space for efficient spatial similarity calculations.

A critical innovation in DeepION was the introduction of modality-specific data augmentations based strictly on MSI physical prior knowledge. Because ion detection follows Poisson statistical distributions and matrix crystallization causes signal dropouts, the model injects calculated Poisson noise and simulates random missing value patterns during training. The "COL" mode is optimized for colocalized ions, while the "ISO" mode utilizes intensity-dependent missing value operations to discover isotope ions. DeepION achieved remarkable spatial identification accuracy on rat brain tissue, drastically eclipsing standard contrastive models and traditional linear dimensionality reduction techniques.

Deep Learning for Spectral Similarity and Structural Annotation

Once peaks are autonomously extracted and spatially grouped, the final computational hurdle is the structural identification of the molecules those peaks represent. Historically, the "Cosine Score" served as the gold standard for comparing tandem mass spectra against reference libraries. However, cosine similarity suffers from a high false-positive rate and fails to connect structurally similar molecules if minor functional group modifications shift fragment ion masses.

To overcome this, the field has adopted deep representation learning. The Spec2Vec algorithm from [Huber et al. \[2021b\]](#) adapts the Word2Vec NLP architecture to untargeted metabolomics. By training unsupervised on diverse spectra, it maps mass fragments to an abstract, multi-dimensional geometric embedding space. Spectral similarity is calculated based on geometric proximity, demonstrating significantly better correlation with true structural similarity than classical cosine scores.

Building upon this, [Huber et al. \[2021a\]](#) introduces MS2DeepScore which is a highly supervised approach utilizing a deep Siamese neural network. It is mathematically trained to predict the true structural similarity (Tanimoto score) between two molecules directly from their mass spectra. When benchmarking statistical precision across similarity thresholds, MS2DeepScore consistently outperforms both classical cosine metrics and Spec2Vec, allowing for chemically meaningful spatial clustering even when exact library matches are unavailable.

2.3 Summary

Mass Spectrometry Imaging is a transformative technology offering a label-free, detailed window into the biochemical landscapes of intact biological tissues. However, its high-dimensional complexity poses severe computational challenges. Traditional preprocessing pipelines, particularly algorithmic peak picking, dangerously bias biological interpretation by discarding low-abundance signaling molecules and disrupting non-linear spectral manifolds.

The scientific literature demonstrates an accelerating paradigm shift toward deep representation learning architectures to resolve these bottlenecks. Generative models (like msiPL) and spatially-aware extensions (like S3PL) mathematically learn latent spectral manifolds directly from raw data to autonomously identify critical molecular peaks. Self-supervised contrastive networks (like DeepION) utilize physical instrumentation priors to drastically improve the detection of colocalized ions. Finally, structural annotation is now optimized by advanced deep embedding networks, such as Spec2Vec and MS2DeepScore, which model chemical relationships more robustly than classical distance metrics.

Chapter 3

Research Methodology

3.1 Research Questions

This project is guided by the following specific research questions:

1. How can localized spatial covariance be mathematically integrated into the probabilistic encoding process of a standard 1D Variational Autoencoder without introducing the severe computational and memory overhead associated with 3D convolutional networks?
2. Does the application of an explicit spatial coherence penalty combined with physical noise augmentations effectively isolate true biomolecular signals from stochastic matrix noise?
3. To what extent does this spatially regularized framework improve the identification of biologically relevant, spatially structured peaks compared to standard, spatially-blind models like msiPL?

3.2 Research Aims and Objectives

Aim: To adapt and evaluate a Spatially-Aware Variational Autoencoder (Spatial-msiPL) capable of autonomously identifying spatially coherent molecular peaks in untargeted Mass Spectrometry Imaging data, acting as a computationally efficient alternative to 3D convolutional networks.

Objectives:

- Curate and preprocess high-resolution, raw profile-mode MSI datasets from public repositories.
- Formulate a modified VAE architecture that ingests contextual input vectors (concatenating central spectra with spatial neighborhood averages).

- Augment the standard Evidence Lower Bound (ELBO) loss function with a spatial penalty term to enforce biological coherence.
- Extract relevant mass-to-charge (m/z) peaks autonomously using back-propagated mathematical weight analysis.
- Quantitatively evaluate the extracted ion images against expert-annotated structural masks using the mean Spatial F1-score (mSCF1) and Pearson Correlation Coefficients.

3.3 Dataset(s)

To ensure strict scientific reproducibility, the proposed model will be trained and evaluated against diverse, high-resolution public datasets sourced from the ProteomeXchange consortium and the METASPACE knowledge base. A strict methodological requirement for this project is the exclusive utilization of raw, profile-mode mass spectrometry data stored in the standard imzML format. Centroided data strips the non-linear morphological wave-shapes that representation learning models require to differentiate signal from noise.

Specific benchmark datasets selected for this study include:

- **Glioblastoma Multiforme (GBM) Dataset:** A highly heterogeneous brain tumor model sourced via the National Metabolomics Data Repository (NMDR) under project ID PR001292.[[Abdelmoula et al. 2021](#)]
- **Human Colorectal Adenocarcinoma (CAC) Dataset:** Available from the MetaboLights repository under accession MTBLS415.

Expert-annotated ground truth segmentation masks corresponding to these datasets will be sourced from the open-source S3PL benchmark repository.[[Weigand et al. 2026](#)]

3.4 Preprocessing

The datasets will undergo Total-Ion-Count (TIC) normalization. To introduce spatial awareness before network ingestion, the input space will be restructured. For any given spatial coordinate, a contextual vector $\tilde{x}^{(i)}$ will be generated by mathematically concatenating the original central spectrum $x^{(i)}$ with the spatially averaged spectra of its immediate 8-pixel Moore neighborhood. Furthermore, inspired by DeepION, synthetic Poisson statistical noise will be injected into the input arrays to simulate the true physical ion detection

mechanics of the mass spectrometer, preventing the model from overfitting to localized matrix artifacts.[Weigand et al. 2026]

3.5 Models and Algorithms Used

The core model, **Spatial-msiPL**, builds upon the proven 1D Variational Autoencoder architecture. The probabilistic encoder network infers the distribution of a low-dimensional latent variable z from the contextual vector. The decoder network then attempts to accurately reconstruct *only* the original, isolated center spectrum. This architectural bottleneck forces the network to utilize the broader spatial context to denoise the localized output.

The network will be optimized by maximizing the Evidence Lower Bound (ELBO), augmented by a novel explicit spatial regularization term, $\lambda \mathcal{L}_{\text{spatial}}$. This penalty calculates the Euclidean distance between the latent representation of a central pixel and the mean latent representation of its physical neighbors, smoothing the latent manifold:

Following successful convergence, unsupervised peak extraction will be executed via a backpropagated analysis on the connection weights $W^{(1)}$ linking the input layer to the first hidden layer. The relative importance of each specific m/z bin is determined by calculating the $L2$ -norm across all hidden units, bypassing heuristic thresholds entirely.[Weigand et al. 2026]

3.6 Evaluation Method

Because defining an absolute biological “ground truth” for untargeted peak picking is exceptionally difficult across tens of thousands of ions, the evaluation procedure will utilize the expert-annotated structural segmentation masks.

We will benchmark Spatial-msiPL against current state-of-the-art peak picking methods (standard msiPL, S3PL, and MALDIquant). Performance will be evaluated by computing the Pearson Correlation Coefficient (PCC) between the algorithmically extracted ion images and the pathological binary masks. The overall precision and recall of the spatial structure will be determined using the mean Spatial F1-score (mSCF1).[Weigand et al. 2026] This ensures the model is objectively evaluated on its ability to pick biologically relevant spatial structures rather than isolated noise.

Chapter 4

Work Schedule

The project will be executed over a 4-month period, divided into the following phases:

- **Month 1 (Literature & Data):** Consolidate the literature review, navigate public repositories, and acquire/parse the multi-gigabyte raw imzML profile datasets (GBM and CAC) and their corresponding segmentation masks.
- **Month 2 (Preprocessing & Baseline):** Implement the data preprocessing pipeline (TIC normalization, contextual vector generation) and replicate the baseline msiPL model for comparative analysis.
- **Month 3 (Model Development):** Architect the Spatial-msiPL network, integrate the spatial coherence penalty into the ELBO loss function, and conduct training with Poisson noise augmentations.
- **Month 4 (Evaluation & Documentation):** Extract final peak lists, execute comparative benchmarking using mSCF1 and PCC metrics, and compile the final Honours research report.

Chapter 5

Conclusion

The computational bottleneck of heuristic peak picking remains a primary barrier preventing the clinical automation of Mass Spectrometry Imaging. While unsupervised spectral generative models like the foundational msiPL framework have proven the viability of mathematical feature extraction, their failure to utilize biological spatial context limits their accuracy. The proposed Spatial-msiPL architecture seeks to bridge this critical gap by mathematically integrating physical neighborhood covariance directly into the encoding process. By evaluating this hybrid model using rigorous spatial correlation metrics, this research aims to contribute a highly efficient, spatially-aware deep learning framework capable of autonomously mapping the biochemical topography of diseased tissues without the severe computational overhead of full 3D convolutions.

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